

The Morning Newspaper

MONDAY, SEPTEMBER 28, 2026 · AUSTIN EDITION

97° / 76°

Overcast, 4% rain, wind 12 mph

Tomorrow: 93° / 77°, overcast

Sunrise 7:22 · Sunset 19:20

Waning gibbous

Iran Talks Collapse, Treasury Yields Hit Decades High as Wall Street Slides

President Trump rebuffed Iran's latest peace proposal Monday, sending oil higher and 10-year Treasury yields to their highest level since 2007 as stocks logged a third straight session of losses.

WORLD & TECH

FAA Delays Boeing 737 MAX 10 Certification

REUTERS

The Federal Aviation Administration said Monday it will not certify Boeing's 737 MAX 10 until it evaluates a newly discovered software glitch affecting the jet's navigation systems. The news sent Boeing shares down nearly 7 percent, their worst single-day drop in months and the biggest drag on the Dow. Airlines that had been counting on the larger jet to replace aging fleets now face another year of delivery uncertainty.

OpenAI Shelves Its Next Flagship Model

AI AGENTS DIRECTORY

OpenAI has abandoned plans to release its next major model after internal safety reviewers raised fresh concerns, according to industry reports Monday. The company gave no timeline for when, or whether, a revised version might ship, leaving rivals an opening to claim the news cycle while OpenAI's engineers rework the training pipeline.

Nvidia Approves Record \$150 Billion Buyback

CNBC

Nvidia's board authorized an additional \$150 billion in share repurchases, lifting its total

buyback authorization to \$235 billion, the largest ever announced by a U.S. company. The move came even as chipmakers broadly slid with the rest of the tech sector Monday, a sign the board sees the pullback as sentiment-driven rather than a break in AI demand.

MARKETS

INDEX	LAST	CHG
S&P 500	7,683.69	-0.77%
Dow Jones	51,481.51	-0.67%
Nasdaq	26,820.38	-0.92%
Russell 2000	2,817.91	-0.69%
VIX	16.07	+8.07%
10-yr yield	5.24	+1.08%
CRYPTO		
Bitcoin	\$83,282	-1.35%
Ethereum	\$2,680.43	-0.11%
STOCKS		
NVDA	\$228.86	+1.68%
AAPL	\$338.40	-0.78%
MSFT	\$509.22	-1.35%
GOOGL	\$342.75	-0.34%
AMZN	\$246.15	-1.41%
META	\$715.62	-4.79%
AVGO	\$349.57	-0.92%
TSLA	\$357.45	-3.94%
SPCX	\$145.47	-2.16%
BRK-B	\$503.09	-0.47%
TSM	\$452.88	+0.50%

Stocks fell for a third straight session as the 10-year Treasury yield climbed to 5.24%, its highest since 2007, reviving fears the Fed will need to keep rates higher for longer. Renewed U.S.-Iran tensions pushed oil prices up and rattled risk appetite, while Boeing's near-7% slide after an FAA certification delay weighed heavily on the Dow. Nvidia bucked the gloom after its board approved a record \$150 billion buyback, though the broader chip sector still finished lower. Bitcoin and Ether slipped in step with equities, extending a rough 24 hours for crypto. Small caps and rate-sensitive names in the Russell 2000 took the sharpest hit, and the VIX jumped more than 8% as traders paid up for downside protection heading into a data-heavy week.

ON THIS DAY

1928: Bacteriologist Alexander Fleming returns from vacation to find a stray mold has killed the bacteria in a petri dish he'd left out, the accidental discovery of penicillin. It would take over a decade, and a wartime push by Florey and Chain, before the drug reached patients, but the antibiotic era had begun.

FROM THE FEEDS · FULL TEXT IN SECTION B

AEON

Paternity is poetical — Daniel Swift argues fatherhood and poetic creativity aren't at odds, and neuroscience backs him up.

LONGREADS

Oh, How Lucky We Are — a Georgia island's last steward and the case for embracing what we can't understand.

LONGREADS

A Graveyard Shift with Britain's Gen Z Undertakers — could apprentice morticians solve youth unemployment?

NAUTILUS

Fire in the Brain — when the immune system attacks brain cells, the result can look like psychosis.

NAUTILUS

Does Religion Change Our Brains? — devotional practice appears to reshape the brain in measurable ways.

PSYCHE

Can He Teach the Teachers? — Stanislas Dehaene wants classrooms built on brain science, but politics gets in the way.

QUANTA MAGAZINE

Mathematicians Crack a 55-Year-Old Conjecture — a juggling-inspired number puzzle finally falls to probability.

TRICYCLE

Beauty, Being, and Becoming — what flowers can teach about gratitude, generosity, and letting go.

LONGREADS

Oh, How Lucky We Are

KRISTA STEVENS · LONGREADS.COM · SEP 28, 2026

A rich profile of Sandy West and her decades stewarding Georgia's Ossabaw Island.

Beryl Gilothwest's grandmother, Sandy "Moose" West, passed away at age 108 in 2021. For *The Bitter Southerner*, Gilothwest profiles a diminutive woman whose love for Ossabaw, an island off the coast of Georgia, was larger than life. In a world striving for "progress," West was a staunch advocate for thoughtful preservation; she was a fierce conservationist and environmentalist who embraced the unknown and took pleasure in learning from it.

"Sandy believed that embracing what we cannot understand is the pathway to enlightenment. In her mind, spending time in the uncharted wilds of Ossabaw is the best way to learn that lesson. Separated from the mainland by a 30-minute boat ride, the island is deeply remote. Cell service is scant, and there is no wi-fi. You have

to fend for yourself out there in a way that is uncommon in contemporary culture, where we always seem to have an escape hatch. Sandy knew that welcoming people into that kind of environment, where they have to relinquish control to the island, would force them to reckon with the unknown."

More picks from *The Bitter Southerner*

"With little fanfare, Ray Di Pietro has attracted tens of thousands of devotees. His images are artfully composed, spontaneous, honest, and, well, kind."

"The fact that I had song ideas, let alone a place where I was saving them, was significant. It had been 15 years since I'd last released an album."

AEON | A WORLD OF IDEAS

Paternity is poetical

DANIEL SWIFT · AEON.CO · SEP 28, 2026

An essay backed by neuroscience arguing fatherhood fuels rather than blunts creativity.

Here's an old story, often told. In Book 6 of the *Iliad*, Hector – dressed splendidly in bronze armour and a helmet with a horsehair crest – bids farewell to his wife Andromache. She begs him not to return to battle and hands him their newborn son. Hector raises the boy into the air, says a short prayer, and hands him back. 'Go home, attend to your own handiwork / at loom and spindle,' he commands, in Robert Fitzgerald's translation: 'As for the war, / that is for men.'

Hector is a father in the heroic mode, and his parenting style has long been irresistible to artists and writers. Few were shocked when the painter Paul Gauguin left his wife and five children in Denmark to pursue fame and art in Paris, then Tahiti. Such distance, they say, is good for a child. 'The father, from his distance, supports, protects, nourishes his child,' wrote D H Lawrence in 1922: 'and it is ultimately on the remote but powerful father-love that the infant rests.' Distance is equally necessary for the father who seeks to make art. 'There is no more sombre enemy of good art than the pram in the hall,' wrote Cyril Connolly in *Enemies of Promise* (1938).

This is a deeply self-serving story, of course. It's nice for the male poet not to have to change diapers. But the argument that unsullied privacy is essential to creativity is tenacious. It surfaces when Virginia Woolf writes that 'a woman must have money and a room of her own if she is to write fiction.' Contemporary feminists have continued to respond to this archetype of generativity. In Jenny Offill's novel *Dept. of Speculation* (2014), for example, the narrator vows never to marry but to become 'an art monster instead'.

But what if domesticity – and specifically parenthood – is in fact generative for artists? What if the cliché of solitary male creativity

got it wrong? We may begin with one nursing poet. In his poem 'Frost at Midnight', written in the winter of 1797, Samuel Taylor Coleridge pictures himself at home in his cottage on a frosty night. The owls cry outside and the frost rings the trees and he is alone, 'save that at my side / My cradled infant slumbers peacefully.' He addresses his infant son:

Dear Babe, that sleepest cradled by my side,

Whose gentle breathings, heard in this deep calm,

Fill up the intersperséd vacancies

And momentary pauses of the thought!

He is transported to memories of his own childhood, and then his hopes for his son's future.

'Frost at Midnight' is a poem about caring, in both senses of the word: caring for and caring about. As such, it is a surprising poem about domestic labour, which has invited comment from scholars. 'What is truly radical about "Frost at Midnight" is Coleridge's self-representation as a father in the traditional maternal posture of watching over a sleeping baby,' notes Jonathan Bate. What Bate calls 'the traditional maternal posture' is what Hector calls 'your own handiwork / at loom and spindle'. But perhaps it is not so radical to picture a poet who is equally involved in the work of parenting. Perhaps it is precisely fatherhood that made Coleridge a poet.

What makes 'Frost at Midnight' significant in Coleridge's career is that it represents the crest of the much longer wave of his stylistic development. It is not just a great poem – it is the poem within which he becomes a great poet, and his growth as a poet can be tracked neatly against his experience of becoming a father.

Coleridge was far from home when his son was born. A lost revolutionary, he was always trying to save or at least improve the world, and he was preaching in Birmingham when word came by letter: Hartley, born to his wife Sara, on 19 September 1796. He met the news with shock and a trio of sonnets. ‘My o’erwhelmed Heart / Trembled, & vacant tears stream’d down my face,’ he wrote in the first of these. He sent this to a friend, and it was published only in 1847, after his death. He wrote another on the journey home, and here imagines that he might arrive to terrible news: ‘when I reach my door, / If heavy looks should tell me thou art dead ...’ They are not charming poems, but repressed; so guarded even when apparently tender, as if he is trying out how he feels in the set form of an Italian sonnet. ‘When I saw it on its mother’s arm, / And hanging at her bosom,’ he confesses in the last of these sonnets: ‘Then I was thrill’d and melted.’ This is a cautious, half-convincing expression of fatherly love. A few days later, he wrote to a friend: ‘in truth, I have seen handsomer babies in my lifetime.’

‘Kubla Khan’, ‘Christabel’, ‘The Rime of the Ancient Mariner’ were all begun or completed within 18 months of his son’s birth

Because Coleridge narrated his moods, and wrote letters that turned into poems, he permits us to reconstruct a detailed timeline of his

changing feelings towards his son. Later, it will be an unhappy story, and Hartley will live an unhappy life, but at first it did not seem to be going that way. By December 1796, Coleridge was planning to move his family to a thatched cottage in the village of Nether Stowey in Somerset. He wrote to a friend with enthusiasm: ‘I mean to work very hard as Cook, Butler, Scullion, Shoe-cleaner, occasional Nurse.’ He turned out to love nursing. By the start of February, he was writing to another friend: ‘You would smile to see my eye rolling up to the ceiling in a Lyric fury, and on my knee a Diaper pinned, to warm.’ This moment – of writing while nursing, of diapers and poetic inspiration – is the genesis of ‘Frost at Midnight’.

The creation of the father is also the creation of the poet. In the spring of 1797, Coleridge was writing 50 lines of blank verse a day. He began to develop his ‘conversation poems’, which drew from his letters to friends and family: their language is intimate, domestic; their topics often autobiographical. Towards the end of that year, he turned to what are now called his ‘mystery poems’, and are those for which he is most celebrated: ‘Kubla Khan’, ‘Christabel’, ‘The Rime of the Ancient Mariner’, all of which were begun or completed within 18 months of Hartley’s birth.



Continued online (full article about 3882 words). Scan, or visit <https://aeon.co/essays/when-a-man-becomes-a-father-and-a-poetic-revolution-follows>

LONGREADS

A Graveyard Shift with Britain's Gen Z Undertakers

CHERI LUCAS ROWLANDS · LONGREADS.COM · SEP 28, 2026

On-the-ground reporting on young apprentices entering the funeral trade.

In this story for Dispatch, Lotte Brundle reports on a surprising trend: young women in the UK are pursuing careers in the death care industry. In a time of high unemployment, funeral home apprenticeships offer an alternative to university, where students accumulate significant debt. While the pay is low and the industry needs to do a better job building awareness that the trajectory exists, the young women who speak to Brundle find the work meaningful and rewarding.

Her inspiration to become an undertaker arrived, as many teenage epiphanies now do, via TikTok. Emily was 15 and had been considering becoming a teacher or journalist when she came across

an American woman documenting life inside her own funeral home. ‘I’d never considered it as a career,’ she says. ‘I didn’t know anyone who was an undertaker, but I just thought how interesting it was.’

Two hours later, she was still scrolling through the woman’s profile, engrossed. Running her own funeral home became Emily’s dream.

More stories about death

‘In most of the country, executions are a thing of the past. But one state has been carrying them out at a record pace.’

‘After my husband’s death, I had never been more pliable, tender, open, or raw. It was then that I tried E.M.D.R. therapy.’

NAUTILUS

Fire in the Brain: How Psychosis May Arise When the Body Attacks Itself

GEORGIA MICHELMAN · NAUTILUS · SEP 28, 2026

Explains how misdirected antibodies can produce psychiatric symptoms once mistaken for mental illness alone.

This story was originally published by Knowable Magazine. Peter Jones once treated a patient with strikingly vivid hallucinations of animals. During ward visits, the patient would stroke and interact with the critters “in as real a way as he was with

me,” says Jones, an academic psychiatrist at the University of Cambridge in England.

Antipsychotic drugs did nothing to dissolve these Snow White-esque visions. But when the patient received plasma exchange — a

treatment normally reserved for a misbehaving immune system — his hallucinations dissipated.

Clearly, this was not a textbook case of schizophrenia. Rather, the patient's symptoms stemmed from autoimmune encephalitis, a rare form of brain inflammation that can, in some cases, cause hallucinations and other symptoms of psychosis. When these patients are treated with drugs that tamp down the immune system, the psychotic symptoms often disappear.

Autoimmune encephalitis is usually first identified through testing blood or spinal fluid for certain types of antibodies. If patients test positive for these antibodies, it's a sign that their immune system may be attacking the brain. Other signs can help to solidify diagnosis: bright patches of inflammation on MRI scans, quickly worsening psychotic symptoms and — importantly — clear neurological symptoms such as seizures.

But Jones' patient didn't have those obvious neurological symptoms. If Jones hadn't suspected an autoimmune cause, the case likely would have been missed.

Some experts suspect that the field of psychiatry is overlooking a small number of patients like this, and they are working to find more of them. They're considering screening a broader array of psychiatric patients and seeking out new antibodies that may be involved in autoimmune encephalitis.

At the same time, many worry that there's a risk of planting false hopes and diverting patients from the psychiatric treatments the vast majority will actually need, should autoimmune encephalitis become overdiagnosed.

Landmark Research Paper

The "artificial divide" between the body and the mind that separates the fields of psychiatry and neurology is starting to break down, says Michael Zandi, a neurologist at University College London. Psychosis, for example, can be caused by a mix of mental and biological factors, from psychological trauma to drug abuse to physical diseases or infections (such as syphilis) that alter brain activity. Around 3 percent of people will experience a psychotic episode in their lifetime.

Brain inflammation caused by the body assaulting itself, as in autoimmune encephalitis, has been recognized for the past couple of decades. In the early stages of this condition, many patients develop the core features of psychosis — hallucinations, delusions and agitation — and so are initially sent to psychiatric services.

A landmark paper about this condition came in 2007, when researchers described the cases of 12 women with autoimmune encephalitis. Nine of the patients initially presented with a number of psychosis-like symptoms, while the others mainly had memory loss. All but one devolved into seizures. The researchers found that the women's conditions were caused by antibodies that bind to the brain's NMDA receptors, proteins that are involved in many processes in the brain, such as learning and the formation of memories.

The trigger for autoimmune encephalitis, in this case, seemed to be tumors in these patients, 90 percent of which were ovarian. All five tumors the researchers examined contained nerve tissue complete with NMDA receptors. This confused the immune system, which

came to recognize the NMDA receptor as foreign and ramped up the manufacture of antibodies that bind to it.

But those antibodies also bound to NMDA receptors in healthy brain tissue, interfering with brain function. Other cells of the immune system gathered to the area, sparking inflammation that, in many of these cases, caused psychosis.

Beyond tumors, experts also think that classic autoimmune encephalitis may be triggered by previous infections such as herpes simplex virus in the brain. But root causes are often mysterious. Only a tiny sliver of psychosis cases are due to autoimmune causes — most experts cap it at 1 to 2 percent of patients. If untreated, these patients may decline into a coma and even die, so to avoid missing them, savvy psychiatrists look for other neurological symptoms, most notably seizures, but also features like catatonia or mouth-twitching.

If these telltale signs of brain inflammation emerge, blood or spinal fluid may be tested for autoantibodies (antibodies that attack the patient's own tissues). If these tests, often coupled with signs on MRI or EEG brain scans, point to autoimmune encephalitis, patients are rushed into immunotherapy: treatments with drugs to tamp down the overreactive immune system. Many fully recover: "We can get them completely better," says Zandi.

Autoimmune encephalitis is commonly caused by antibodies that bind to the NMDA receptor, known as NMDAR antibodies. But researchers have uncovered dozens more antibody types that are associated with encephalitis, some of which also appear to produce psychiatric symptoms.

One leading example is LGI1 antibodies, which were discovered in 2010. These antibodies latch onto LGI1 proteins in the brain, preventing them from passing signals between nerve cells. This form of autoimmune encephalitis often causes dementia-like symptoms, leading to initial misdiagnosis. Zandi says he's had patients whose apparent dementia has disappeared after immunotherapy, leaving them "fully independent, back to driving, normal."

University of Oxford psychiatrist Belinda Lennox has observed similarly striking transformations. Around the time of the 2007 paper, she began conducting antibody tests on psychiatric patients whose symptoms seemed to overlap with signs of autoimmune encephalitis, such as quickly shifting psychotic symptoms or neurological features like seizures.

One of Lennox's first patients to test positive for NMDAR antibodies was a young female librarian who had rapidly descended into paranoia, stopped sleeping and then went mute. Antipsychotic drugs made her condition worse. After the antibodies were detected, the patient was given treatment to remove them from her blood. Within about a week, she was completely better. "That was the sort of mind-blowing moment," Lennox says.

Today, Lennox is fascinated by the possibility of a larger population of psychosis patients whose illness is owed to autoimmune issues that are missed. This might explain why some patients don't respond well to psychiatric treatments. "It's an unexplored tip of the iceberg," she says.

In search of missed patients, for the last decade Lennox has been running a clinical trial called SINAPPS2 to probe whether antibodies can be causing psychosis even in patients who don't have other

obvious neurological symptoms such as seizures and comas. Jones of the University of Cambridge is another investigator on the trial.



Continued online (full article about 2873 words). Scan, or visit <https://nautil.us/fire-in-the-brain-how-psychosis-may-arise-when-the-body-attacks-itself-1285334/>

NAUTILUS

Does Religion Change Our Brains?

KRISTEN FRENCH · NAUTIL.US · SEP 28, 2026

Surveys evidence that devotional practice leaves measurable marks on brain structure.

From the age of three, the 12th Century nun, mystic, poet and composer Hildegard von Bingen had dazzling visions of shimmering points of light and falling stars, which she recorded in detail in her writings and in paintings. The visions were accompanied by revelations about scripture and humanity’s place in the cosmos that she attributed to the will of God.

Neurologist Oliver Sacks famously argued in his 1970 book *Migraine* that Von Bingen’s visions were “indisputably” migraine auras, echoing an earlier hypothesis by British historian of science Charles Singer, who recognized clinical descriptions of visual disturbance in her art. It is impossible to know whether their shared hypothesis was correct. But it has become evident in recent decades, through the emergence of a science sometimes referred to as neurotheology, that the agonies and ecstasies of religious experience are often reflected in the workings and architecture of our brains.

Read more: “Was Jesus a Shaman?”

Recently, an international team of scientists from Europe, Asia, and the Middle East set out to systematically review how different religious practices show up in the brain. Across the millennia, humans have devised many ways to commune with the divine: meditation, chanting, recitation, musical worship, prayer, mediumistic trance. Some of these practices require more devotional attention than others, and some of them can produce altered states of consciousness. The team synthesized the findings of 105 studies spanning 28 years (1998-2026), across eight different religious traditions, and seven different types of imaging.

Their primary finding was that religious and spiritual practices can produce consistent, and in some cases, lasting changes to how the physical brain looks and works, such as altered gray matter or connectivity between regions. Each practice differed in significant ways from the next, but two regions showed up most frequently in the results. The first was the prefrontal cortex, the brain’s control center, which handles high-level thinking, decision-making, and emotional control. The second was the anterior cingulate cortex, deep in the middle of the brain, which acts as a bridge between the brain’s thinking areas and emotional areas. They published their results in the journal *Brain and Behavior*.

Devotional OM chanting, for example, seems to calm both the brain’s emotional alarm center, the amygdala, and other emotion centers found deep in the limbic system, the team found. But Buddhist

chanting activated the amygdala instead, and seemed to tune the dial on other emotional centers in the limbic system. Likewise, how prayer shaped the brain seemed to depend very much on the type of prayer practiced. Improvised personal prayer was found to activate regions of the brain central to interpersonal connection. Another kind of prayer associated with Sahaja yoga, aimed at quieting the mind, forgiveness, and humility, seemed to deactivate certain parts of the brain associated with external awareness and top-down mental control. And people regularly practicing a type of daily Islamic prayer called Salat, which includes posture, recitation, breathing, and focus, showed differences in the corpus callosum, which connects the two cerebral hemispheres of the brain.

Mediumistic trance, meanwhile, as measured by electroencephalogram, looked like a distinct and healthy form of consciousness and produced specific patterns in slower brain waves associated with a person entering a deeply absorbed, automatic state like flow. Some results were counter-intuitive. Hindu Vedic priests trained from a young age in memorization and recitation of thousands of Sanskrit stanzas showed significantly greater cortical thickness than matched controls, localized to regions of the brain involved in self-discipline and emotional regulation, as well as pitch, rhythm, and melody. But they did not show changes in the hippocampus, the brain’s hub for forming new memories.

While most of the study participants were healthy, a small subset of them suffered from mental health conditions such as depression. The authors suggest that the findings from these studies yield insights helpful in closing the gap in global mental health treatment. Religious and spiritual practices have not yet been reliably shown to treat diagnosed mental health conditions, but they are, they note, “already ubiquitous, freely available, culturally legitimate, and actively engaged in by a substantial proportion of the population.” Many of the studies were also small, correlational, or single case studies, so the findings are preliminary—a starting point.

Hildegard von Bingen never had a brain scan, but nuns who came after her have, and their brains, like those of monks, Vedic priests, and other religious practitioners, show differences that seem to track how they spent their lives. If the causal reading holds, it suggests that humans may have been actively shaping the way their brains work through religious practice since the beginning of time.

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Lead Image: STILLFX / Shutterstock

Mathematicians Harness Randomness To Crack a 55-Year-Old Conjecture

SHALMA WEGSMAN SEPTEMBER · QUANTAMAGAZINE.ORG · SEP 28, 2026

A technical but readable account of how probabilistic methods finally resolved a decades-old rearrangement conjecture.

The late Ronald Graham wore two hats. He was a renowned mathematician, at one time president of the American Mathematical Society. He was also a serious juggler, and president of the International Jugglers' Association. "He loved tricks," said Fan Chung, a mathematician at the University of California, San Diego, who was married to Graham. "You know, spinning a ball, spinning a coat hanger, spinning several balls together, throwing pens against the wall."

Sometimes Graham wore both hats at once. "It's interesting, in fact, that many mathematicians and computer scientists have an interest in juggling," he said in a 1980 television interview. "I think it's the search for patterns and structure that is responsible for this."

He would go on to write numerous papers, some with Chung, on the mathematics of juggling. But back in 1971, decades before he made that connection explicit, he posed a question that some mathematicians now say might have been inspired by juggling, too.

Start with a random set of different integers, not including zero. Can you always rearrange them so that if you add up the first two numbers, then the first three, then the first four, and so on, every "partial sum" turns out different? In the language of juggling, this would mean that if each ball stays in the air for a different amount of time, you can always find an order to throw them in such that two balls won't come crashing down on the same beat — which would ruin the act.

If the numbers are all positive, then the answer to Graham's question is obviously yes: The sums will always grow larger as you add more numbers. Similarly, if there are both positive and negative numbers in the mix, the answer is also known to be yes. But what if the numbers live in a finite world — like numbers wrapped around a clock, which repeat after a certain count?

That's what Graham wanted to know. He conjectured that the answer should still be yes. It often happens, he figured, that even when dealing with rigid constraints, you can still find enough flexibility to construct special patterns or structures — just as it's usually possible to find a valid sudoku board or Latin square (another kind of puzzle) despite their many rules. "It fits nicely in all these questions about designs and about very symmetric structures," said Noga Alon, a mathematician at Princeton University. But for decades, no one could prove Graham's intuition to be true.

That changed recently, when several young mathematicians picked up the balls. In a proof that spanned four papers and various fields of mathematics, they finally resolved Graham's rearrangement conjecture. The final paper, by Lisa Sauermann of the University of Bonn and Huy Tuan Pham of the University of Chicago, appeared in February 2026, officially closing the problem.

Across the papers, one theme prevailed: the power of randomness to draw out patterns. As Alon put it, "It's the power of collaboration, the

power of the young generation, the power of probabilistic methods" that solved the problem.

Alp Müyesser, a mathematician at the University of Oxford, often finds himself drawn to problems whose solutions need two ingredients: a random process, and something extra as well. After solving one such problem in 2022 while he was still a graduate student, he encountered Graham's conjecture and realized that his just-finished proof could help there, too.

The conjecture is set in the world of clock arithmetic. You start by placing the whole numbers on a number line, then you wrap the line around the face of a clock so that the numbers repeat after some prime number, p . Say p is 7, for instance. In this setting, 0, 7, 14, and all other multiples of 7 are equivalent — meaning that you can add two positive numbers (like 3 and 4) and get zero.

Graham asked the following: If you pick any set of nonzero numbers off this number line (for any p), can you always rearrange them so that the partial sums you get are all different?

The challenge depends on how big your set is compared to p . The more numbers you pick, the more sums there are to manage. But if you choose fewer numbers, there will be fewer ways to rearrange them. These different cases inspire different approaches.

Müyesser, along with his former adviser, Alexey Pokrovskiy of University College London, tackled the case where your set includes almost every possible number up to p . With sets this large, it can be extremely hard to construct a valid ordering. But it turned out that starting with a random ordering can bring you most of the way there.

"Computer scientists often call this a 'finding the hay in the haystack' problem," Müyesser said. You might know that lots of good orderings are out there, but actually finding one is hard. "If you do it randomly, it's likely going to work, but it's hard to explicitly describe what the solution is supposed to look like."

Müyesser and Pokrovskiy needed to ensure that no sequence of numbers anywhere in the ordering added up to zero. Otherwise, adding those numbers to the previous partial sum would repeat that sum.

A completely random ordering might have a few of these troublesome sequences. So Müyesser and Pokrovskiy first set aside a few specially chosen numbers from the set, then randomly scrambled the rest. They scanned their random ordering for any problems; if they came across an interval that added up to zero, they could insert one of the spare numbers to change it. In 2022, they posted their solution, though it was hidden in a paper that focused on applying the same technique to a more general problem.

A couple of years later, Noah Kravitz of Oxford, unaware of Müyesser and Pokrovskiy's solution, stumbled on Graham's conjecture in an online archive of unsolved problems. "I saw there was an open problem, and I was like, it's embarrassing for humanity

that we don't know this," Kravitz said. "This situation just had to be rectified."

He decided to approach the conjecture from the opposite end. Together with Benjamin Bedert of Oxford, he considered the case where the set of numbers is tiny compared to p — for instance, Alon said, if you have a set of 100 numbers where p is 1 billion.

Kravitz and Bedert solved Graham's conjecture for those cases and posted their proof in September 2024. Műyesser saw it and reached out, sharing his own work; the three of them (plus two other colleagues) then teamed up to extend Műyesser's original approach.

"It was a pretty unlikely combination of people," Kravitz said. He and Műyesser come from two areas of combinatorics that don't typically collaborate. "Different sections have completely different techniques," he said.

Their paper, which they posted in August 2025, handled more cases where the set of numbers is relatively large compared to p . But between those cases and the small-set cases that Kravitz and Bedert had covered, a gap remained. No one could figure out what to do about medium-size sets, such as those that include roughly half as many numbers as p . "Our methods didn't work there, and there were clear reasons that they would not have worked," Műyesser said.

It seemed as though research on the problem might enter another long hiatus.

Then, in February 2026, a surprise appeared online.

Lisa Sauermann and Huy Tuan Pham were old friends. The two mathematicians had met in 2015 at Stanford University, where Sauermann was a graduate student and Pham an undergraduate. Today they live on different continents — Sauermann in Bonn, Germany, and Pham in Chicago. But a conference in Germany in September 2025 provided a rare chance for them to share a chalkboard again, and afterward Pham followed Sauermann to Bonn for a short visit. All they needed was a problem to work on.

At the conference, they heard two talks on Graham's conjecture by mathematicians who had attempted but failed to bridge the gap. They were intrigued. And as it later turned out, Sauermann had encountered a closely related problem in the International Mathematical Olympiad as a high school student. She solved it correctly, and by the time she finished high school, she'd won a gold medal in the prestigious competition four times. (Most likely, it was Chung who placed the problem on that year's exam, as she was on the committee that wrote the questions, and she frequently took inspiration from Graham's many puzzles.)

By the end of their three-day visit, Sauermann and Pham had a plan for how to crack the case.

It hinged on a technically demanding method called anti-concentration. Here, an anti-concentration statement asserts that some event has a particularly low chance of happening. But the mechanics of proving these kinds of statements are so intricate that, though Kravitz and others were aware that such an anti-concentration

approach might succeed, "we just hadn't had the guts to actually try it," he said.

First, though, Sauermann and Pham began the way their predecessors had. They randomly reordered their set of numbers and came up with a procedure to fix any problems — that is, any sequences that add up to zero. Any time they found a zero-sum sequence, they swapped out the last number in the sequence with another one.

This procedure often went without a hitch. But three types of "bad events" would cause it to fail. One: A zero-sum sequence might occur toward the end of the entire arrangement; then there would be no other numbers to swap in. Two: Many zero-sum sequences might appear too close together, making it impossible to fix them all. And three: Fixing one bad sequence might create another zero-sum sequence down the line.

Sauermann and Pham hoped to prove, using anti-concentration, that each of these bad events was sufficiently unlikely. Then there would have to be a way to rearrange the set of numbers to satisfy the conjecture.

To do this, the duo used Fourier analysis — an area of math that lets you rewrite functions as sums of simple waves — to show that in general, when you add up random sets of numbers, no one sum is especially likely to appear. They then used this insight to carefully estimate the probability that each bad event would occur, ultimately showing that the total chance of getting a bad event was less than 100%. That was enough to settle the conjecture.

A few months after their stint in Germany, Sauermann and Pham posted their 27-page proof online. They had shown not only that a satisfactory rearrangement was always possible, but that a random ordering could be rearranged to eliminate bad events at least 90% of the time — a massive success rate.

The mathematicians who had previously worked on the problem were surprised to see the remaining case closed so quickly. "Their approach is just completely different," Műyesser said.

Together, the four papers prove Graham's conjecture for sets of all sizes. But they all assume that p is very large; though no one has calculated its exact value, think along the lines of 10 raised to the 100th power. To mathematicians, that's fine — the salient point is that you're working in the setting of clock arithmetic. But another aspect of the problem technically remains unsolved — you might still try to resolve the conjecture for all p . And if you want to use the result to choreograph a real juggling routine, you're out of luck: To correspond to such a large p , the routine would have to be much too long.

The proof confirms that even within these strange, limited number settings, "there are some nice structures that always exist," Alon said. You can always achieve some degree of flexibility, shuffling the numbers in your set around to avoid revisiting the same partial sums.

"To pose a good problem is really an art," Chung said. "I think Ron would be extremely happy to see the problem solved."

Stanislas Dehaene enters a small library inside the mansion that houses the French ministry of education. Autumn light filters down from a skylight, illuminating glass-fronted bookshelves holding the *journaux officiels* – the laws, decrees, regulations and rules of the French Republic. Dehaene takes his place and flips open his laptop as members file in.

As president of the Scientific Council for National Education, a body charged with advising the government on evidence-based learning, Dehaene has a pressing concern: *Le Monde* has just accused the Council of racism. As he shares this news, the warmth in his voice disappears, replaced by something sharper.

He asks the group to share his rebuttal on social media because *Le Monde* has declined to print it. 'I find it ridiculous that we have to waste time responding to such things,' he says. 'It's an accusation completely beside the truth.'

A world-renowned cognitive neuroscientist, Dehaene has published hundreds of refereed journal articles and seven books for general readers on reading, mathematics, consciousness and learning. More than a decade ago, he set out to do something that sounded straightforward: take four decades of research mapping how the brain learns, and compel education systems to follow its insights. He thought the evidence would speak for itself. Instead, he has run into the deeply human resistance of politics, identity and belief. In France, more than one in four children arrive at secondary school without a firm grasp of reading or mathematics. Dehaene does not see these as abstract statistics but rather as the cost children pay for that resistance.

'It's been eight years, and the scores of French children are changing a tiny little bit,' he says with a deep sigh. 'It's very, very difficult. The social forces are enormous.'

The previous day, I visited him at his brain-imaging lab, NeuroSpin, in the Parisian suburb of Saclay, where he took me down a hallway known as 'magnet street' to the facility's massive MRI equipment. He paused outside the control room of the newest MRI magnet, an 11.7-tesla scanner, one of the strongest human-rated machines in the world, which displays the brain in finer anatomical detail than ever seen before. A loud mechanical buzz pulsed in the hallway – the vibration of the magnet's electromagnetic coils. 'Do you have a pacemaker or anything like that?' he asked before opening the door.

Inside, researchers huddled around computer screens, and a video monitor displayed a fuzzy image of the tube's interior. It's one thing to get adults to lie down in the noisy, claustrophobic setting of an MRI tube, but Dehaene, along with his wife, the paediatrician Ghislaine Dehaene-Lambertz, has worked hard to make children comfortable in this setting.

That instinct to observe a child closely would shape Dehaene's future work

In one experiment, they put three-month-olds in the tube – the babies had just been fed, wore headphones, and were placed in a specially designed cradle – and played them short sentences, some spoken normally and some played backward, a way to strip out language's

meaning while keeping its raw sound intact. The results, published in *Science* in 2002, showed that the brain's architecture for language – the same left-hemisphere regions active in adult brains – was already in place and functioning at three months. 'Before our work, people barely knew if the cortex was working or not,' Dehaene says.

He met Ghislaine Lambertz in 1986 while both were working in the lab of Jacques Mehler, a pioneer in cognitive psychology. They became part of a group that met at Mehler's apartment for dinners of pasta and good wine, and they married within a year – Dehaene laughs while sharing that, a few months after the ceremony, their first son, Guillaume, was born. Dehaene was just 21 and, while he had landed his first job, he still hadn't defended his dissertation. 'My son was at my PhD defence,' he says. 'So I guess the order is kid, permanent position, PhD.'

After Guillaume, two more sons, David and Olivier, followed. When Olivier was five, his preschool teacher taught him to write his first name, and he proudly signed all of his artwork. Half the time, though, he wrote the letters right to left. The scientist in Dehaene was fascinated; the father was anxious. Was this a sign of incipient dyslexia? Olivier's backward writing soon disappeared on its own. But that instinct to observe a child closely, to take seriously any unusual turns in their learning behaviour that might signal a problem, would shape Dehaene's future work.

So would Ghislaine. Over the years, she and Dehaene have built a body of research tracing the brain's relationship with language from infancy through the first years of reading. Dehaene showed me his wife's lab, opening a closet where she kept tiny caps fitted with electrodes that could be slipped over a baby's head to read brainwaves. 'We divided the field between us,' he said. 'You know, she is going to do babies, and I won't touch it, and I will do adults ... but of course we mixed in all sorts of ways until we joined in the middle.'

After the tour, we headed to his office, which holds several 3D-printed models of his brain – including one that lay in a pool of powdery plaster dust, the lasagna-like folds of the cortex fractured. Dehaene told me that a photojournalist had asked him to pose with it, and he stumbled and dropped it. 'The guy started taking pictures like crazy,' he says. 'Well, that was the end of my brain.'

Dehaene often punctuates his thoughts with a high-pitched laugh, eyes sparkling, dimples flashing. The effect is disarming. He turned 61 this May but looks years younger – a lean, agile man with a bald pate and delicate wire-rimmed glasses who sometimes wears a tiny red rosette on his lapel signifying his membership in the French Legion of Honour.

Former students and colleagues call Dehaene a genius. At a recent conference, Andreas Nieder, a German neurobiologist, told the audience: 'Whenever I talk with him, I have the impression that his brain is ticking along at twice the frequency of mine.' Later, Ghislaine joked that Dehaene's productivity suggests that he must have cloned himself. 'In fact, there are two Stans!'

'I believe in the expansion of the human mind. I believe this is an extraordinary machine. We should realise its full potential'



Continued online (full article about 4425 words). Scan, or visit <https://psyche.co/portraits/he-mapped-the-learning-brain-then-came-the-hard-part>

NAUTILUS

Famed Dancer's Diary Offers Glimpse into Psychosis

JAKE CURRIE · NAUTIL.US · SEP 28, 2026

A rare first-person record of a descent into psychosis, read through a modern clinical lens.

Called the “god of dance,” Vaslav Nijinsky was one of the most widely renowned dancers of the 20th century. He gained fame as a male ballet dancer and choreographer in the celebrated Parisian dance company, Ballets Russes. Unfortunately, his brilliant career was cut short by a diagnosis of schizophrenia. He was institutionalized in 1919 and spent the last three decades of his life in and out of asylums. During a six-week period when he was gripped by psychosis, he kept a diary, which is now the subject of a new study into how mental illness can affect language.

“Language provides a window into someone’s mind,” explained study author Alban Voppel, professor of AI and psychiatry at McGill University in a statement. “Complete book-length texts written during an active psychotic episode are quite rare, giving us a unique opportunity to study language as the illness was unfolding.”

Read more: “Fire in the Brain: How Psychosis May Arise When the Body Attacks Itself”

Voppel and a team of researchers conducted a computational linguistic autopsy of Nijinsky’s diary using tools that measured patterns in word use as well as sentence structure. They wanted to find out if the linguistic pathologies of formal thought disorders (like schizophrenia) manifest in how topics are connected or in the words used to convey them.

To find out, they compared Nijinsky’s writing to a control text from his wife Romola: the forward she wrote when she published a heavily

edited version of his diary in 1936. They chose her writing because the two texts were of similar length, from roughly the same period, and written about similar subjects. Most importantly, Romola had no history of psychiatric illness.

The team found Nijinsky often wrote in circles using the same words repetitively. This had the effect of demonstrating a kind of superficial coherence, when the work as a whole was generally incoherent. “It’s sort of a fake coherence, a mechanical coherence,” Voppel said. “He came back to the same topics and used a lot of the same words, especially ‘I,’ ‘God’ and ‘love.’”

These words represented a kind of structural scaffold for the rest of Nijinsky’s meandering thoughts—the lettuce in his word salad.

According to the team, this suggests that psychosis can affect language at different levels. It’s possible that a person succumbing to acute mental illness can still communicate in a broad narrative structure while their vocabularies become increasingly limited by their disorder. “Nijinsky’s diary is not simply a document of suffering or a record of thought disorder,” the researchers wrote, “it is an unwitting self-portrait of a mind condensing around a fixed, narrow repertoire of words and the concepts they stand for.”

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NAUTILUS

A Dispatch from the Trenches of Algorithmic Living

NICOLE A PRUCHA · NAUTIL.US · SEP 28, 2026

A personal essay on what it feels like to live inside recommendation algorithms.

This excerpt is reprinted with permission from NYU Press. It is adapted from *Power in Listening: The Sounding Out!* Reader.

Sometime in 2021, I opened the Spotify app, navigating my way to a recent playlist I had made featuring Arlo Parks, Brittany Howard, Hozier, Charlotte Day Wilson, Radiohead, and other artists known, at least in part, for their melancholic sounds and senses. Immediately, under the banner “You Might Like,” was a featured playlist titled “sad girl starter pack.”

As is common in the age of targeted ads, privacy agreements, and social media, I felt simultaneously exposed and validated. As someone who surveys the personalized “Daily Mixes” and “Discover Weekly” playlists, searching for new music daily, it seemed like the perfect addition to my growing playlist collection. Here were 75

songs, some by artists I knew and others I didn’t, that at first glance seemed tailored to my taste. But, the longer I stared at my screen, my excitement turned into skepticism.

Read more: “When You Listen to Music, You’re Never Alone”

Something was off. It, like many of Spotify’s mood-based playlists, claimed to be curated based on a sense, as opposed to a genre, decade, or other category marker. However, unlike the application’s popular mood-based playlists (e.g., “Feelin’ Good,” “All the Feels,” “Broken,” “Chill Vibes”), this playlist evoked both gender and affect, in its name alone. Clearly, it was meant to conjure, capture, and attract a specific version of both femininity and sadness through sound. But what was that version? And what did it, supposedly, sound like?

Since first pressing play on Charlotte Day Wilson’s “Work,” one of the featured tracks, Spotify’s “sad girl starter pack” has continued to haunt me. Early into 2023, Spotify launched a global music program titled GLOW, celebrating and promoting LGBTQIA+ artists and creators. This new digital music hub boasts “Sounds from the Community”—playlists featuring artists who identify as LGBTQIA+—as well as a collection of playlists deemed “LGBTQIA+ Favorites.” Among this latter collection is none other than “sad girl starter pack,” which also happens to be the hub’s most-liked playlist, by far, with over 1.3 million likes. Seeing it there, I once again felt attacked, but from a different angle. To a queer woman who still loves listening to slow, sad songs, how could the same playlist that once raised questions about sadness, femininity, and sound manage to provoke new questions about queerness, too?

In addition to being in the GLOW hub, the playlist’s new description also unmistakably invokes queerness. Two years ago, when the playlist was first released, the description read, “Is her favorite color sage green?” A nod at one of the trendy colors that defined late 2021 and 2022, the description feels playful, harmless, and more so a reference to an aesthetic than a gesture toward the rainbow of identities constituting queerness. Later on in 2022, the description remained somewhat obscure, but became more pointed: “Sapphic songs that defined your music taste as ‘yearning.’” The word “sapphic,” derived from the poet Sappho and often used to describe sexual and romantic attraction between women, clearly suggests queerness. And yet the playlist is not explicitly queer, at least not in an identitarian sense. That is, it does not feature exclusively queer artists like the playlists in GLOW’s “Sounds from the Community” section. It is also not exclusively feminine in that the included artists are not all women-identifying. In fact, as a personalized playlist—Spotify’s hybrid between an editorial and an algorithmic playlist—some of its songs change from user to user.

What seems to be the unifying theme of the playlist, then, is a soft, melancholic sense of yearning that is descriptive of both the music in the playlist and the audience clearly targeted by it. According to the packaging, marketing, and evolution of “sad girl starter pack,” that audience appears to be sad girls, women, and femmes, queer or not. But this collapsing of identities into a single sense, category, or even playlist can be a dangerous game.

Spotify: The Sad Girl’s Digital Domain

As a digital platform, Spotify clearly understands and capitalizes on the sociality and circulatory nature of music. The “Friend Activity” column on the desktop application allows speculation about our friends’ moods based on what they’re listening to; Spotify’s annually anticipated marketing campaign “Wrapped” is designed for sharing across social media platforms with ease; and the app’s “Blend” feature can create a personalized playlist combining up to ten friends’ music tastes. These features underscore something we—music lovers, listeners, and consumers—have always known: Listening is done in, for, and about community. Playlists, arguably Spotify’s defining feature, are thus powerful social tools of relation, no matter their curator. To some, they are epistolary, sent between crushes like vaguely opaque love letters. To others, they are time capsules, relatives of the cassette tape and CD mixtapes that conjure memories and stories. For still others, they are the sonic backdrops of study sessions, breakups, warm-ups, parties, and more. Regardless of how they’re perceived, playlists are inherently relational and deeply affective. It is no wonder then, that “sad girl starter pack” is as popular as it is. Not only does it implicate multiple communities in its title and description, it invites listeners from those communities and beyond to listen, feel, and—to use the language of the playlist—yearn together. But, as this chapter’s opening foreshadows, this rosy depiction of the playlist’s social potential, through affect, begins to unravel as we consider who is hailed by this playlist and how.



Continued online (full article about 2646 words). Scan, or visit <https://nautil.us/a-dispatch-from-the-trenches-of-algorithmic-living-1285345/>

TRICYCLE: THE BUDDHIST REVIEW

Beauty, Being, and Becoming

VALERIE BROWN · TRICYCLE.ORG · SEP 28, 2026

A reflective essay on what flowers teach about gratitude and letting go.

Flowers are the prayers of the vegetal world. The plants offer their beauty to the Supreme.

—The Mother, Flowers and Their Messages

All my life I’ve been a sucker for beauty, especially beauty in the form of flowers, plants, and the natural world. I would rather go without food than without flowers. It sounds dramatic—and that’s

real for me. I recall a time years ago when I was desperate to touch the natural world. I was living in a third-floor walk-up hall room in Queens, working at Burger King, and going to City University of New York in Manhattan at night. On my way to school one day, I passed a street vendor selling roses by the bundle in colors of peach, scarlet, and magenta. Time stopped as I stared at them. I heard them whisper two words to every part of my body: “Take me.”



Continued online (full article about 1248 words). Scan, or visit <https://tricycle.org/article/valerie-brown-flowers-reflection/>